Asymmetric Information and Market Efficiency

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Issues

Accurate asset prices matter for capital allocation (e.g., AI infrastructure boom)

1. Do asset prices incorporate relevant information?

2. How do market participants trade on their information?

Start with question 2, which will allow us to analyze question 1

Road Map

Rational Expectations Equilibrium

Market Efficiency

Model

- $t = 0, 1$
- One risky asset
  - $t = 0$: trades at (endogenous) price p
  - $t = 1$: pays off risky dividend d
- Risk-free asset
  - Exogenous interest rate $r = 0$
- Agents $i = 1, \ldots, N$
  - $t = 0$: endowed with $\overline{n}_i$ units of the risky asset; can trade the risky asset and the risk-free asset
  - $t = 1$: consume value of asset portfolio
  - Risk neutral over date 1 consumption

Agents' Problem

- Agent i maximizes $E[c_i]$ subject to

  date 0 budget constraint: $n_i p + b_i \leq \overline{n}_i p$

  date 1 budget constraint: $c_i \leq n_i d + b_i$

$\Leftrightarrow \max_{n_i} \overline{n}_i p + n_i (E[d] - p)$

Q. What is the equilibrium price? $p = 0$ $p = d$ $p = E[d]$

Equilibrium with Symmetric Information

- Equilibrium price: $p = E[d]$
- Otherwise, demand for the risky asset would be infinite

$$n_i = +\infty \text{ if } p < E[d]$$
$$n_i = -\infty \text{ if } p > E[d]$$

Private Information

- Dividend is uncertain: $d \sim \mathcal{N}\left(\overline{d}, \sigma_d^2\right)$
- At date 0, agent $i = 1$ obtains a signal s about d
  - $s = d + \epsilon$ where $\epsilon \sim \mathcal{N}\left(0, \sigma_\epsilon^2\right)$ and $Cov(\epsilon, d) = 0$

    e.g., data used to predict d (imperfectly)
  - σ_ϵ^2 parameterizes the noisiness of the signal
- Statistics reminder: If d and s are jointly normally distributed, then $E[d \mid s] = E[d] + \dfrac{Cov(d, s)}{Var(s)}\left(s - E[s]\right)$

$\Rightarrow$ The informed agent's posterior belief is

$$E[d \mid s] = \overline{d} + \frac{\sigma_d^2}{\sigma_d^2 + \sigma_\epsilon^2}\left(s - \overline{d}\right)$$

Equilibrium with Private Information

- Suppose agent $i = 1$ receives a positive signal: $s > \overline{d}$

Q1. What does the informed agent ($i = 1$) do in equilibrium?

buy the risky asset sell the risky asset do not trade

Equilibrium with Private Information

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- ..... equilibrium
  - $p = E[d] = \overline{d}$
  - Informed agent ($i = 1$) with posterior belief $E[d \mid s] > \overline{d}$ buys
  - Uninformed agents ($i = 2, \ldots, N$) sell

Q2. Do you find this equilibrium realistic?

Equilibrium with Private Information

- Suppose agent $i = 1$ receives a positive signal: $s > \overline{d}$

Q1. What does the informed agent ($i = 1$) do in equilibrium?

- <u>Naive equilibrium</u>
  - $p = E[d] = \overline{d}$
  - Informed agent ($i = 1$) with posterior belief $E[d \mid s] > \overline{d}$ buys
  - Uninformed agents ($i = 2, \ldots, N$) sell

Q2. Do you find this equilibrium realistic?

- The uninformed agents are naive: they could figure out the informed agent buys because she knows the asset is undervalued $\Rightarrow$ they should not sell

Equilibrium with Private Information

- Less naive equilibrium
  - $p = \dfrac{E[d] + E[d \mid s]}{2}$
  - Informed agent buys
  - Uninformed agents sell

Q3. Do you find this equilibrium realistic?

Equilibrium with Private Information

- Less naive equilibrium
  - $p = \dfrac{E[d] + E[d \mid s]}{2}$
  - Informed agent buys
  - Uninformed agents sell

Q3. Do you find this equilibrium realistic?

- The uninformed agents are still naive: they should not sell

Adverse Selection

- **Adverse selection** arises when some agents have private information
- An agent who trades without taking into account that other agents have private information will systematically buy over-priced assets and sell under-priced assets
- A pervasive issue
  - Financial markets, banking, insurance
  - Any transaction where one side knows more than the other

Equilibrium with Private Information

- Suppose you are an uninformed agent. The asset is trading at $p = 1.2 \times \overline{d}$ and the informed agent is buying

Q. What should you do?

buy sell don't trade

Equilibrium with Private Information

- Suppose you are an uninformed agent. The asset is trading at $p = 1.2 \times \overline{d}$ and the informed agent is buying

Q. What should you do?

buy

Equilibrium with Private Information

- Uninformed agents with rational expectations should update their beliefs when they see the informed agent buying

- Agents have **rational expectations** if their expectations correspond to the true distribution of random variables conditional on their information set (Muth 1961)

Rational Expectations Equilibrium

Q. What is the equilibrium price if agents have rational expectations?

- $p = E[d] \quad E[d|s] \quad$ between $E[d]$ and $E[d|s]$

Rational Expectations Equilibrium

Q. What is the equilibrium price if agents have rational expectations?

- $p = E[d \mid s]$
- The price fully reveals the informed agent's signal
  - Otherwise, the informed agent could trade on her signal and fool the other agents, which cannot happen if they have rational expectations
  - Formal proof below

Rational Expectations Equilibrium

- A Rational Expectations Equilibrium (REE) is described by a price function $P(s)$ which maps agent 1's signal to the risky asset price, such that $p = P(s)$ is an equilibrium when agents have rational expectations
  - Informed agent optimizes using information (s, p)
  - Uninformed agents optimize using information p
  - Market clears
- We can show that
  1. $P(s) = E[d \mid s]$ is the only REE
  2. The price perfectly reveals the private information (mathematically: $P(s)$ is invertible)
  3. Agents don't trade in equilibrium

Proof

2. The price perfectly reveals s
   - $p = E[d \mid s] = \overline{d} + \frac{\sigma_d^2}{\sigma_d^2 + \sigma_\epsilon^2}\left(s - \overline{d}\right)$ is strictly increasing in s, so s can be backed out from p

3. Agents find it (weakly) optimal not to trade
   - All agents have the same posterior belief $E[d \mid s]$, which is equal to the price
   - Therefore, agents are indifferent between any n_i
   - NB: If there are infinitesimal trading costs, agents strictly prefer not to trade

Proof

1. $P(s) = E[d \mid s]$ is a REE

   - Agents optimize: no trade is (weakly) optimal
   - The asset market clears: because agents do not trade

1. $P(s) = E[d \mid s]$ is the only REE

   - By contradiction, suppose there exists another REE $\widehat{P}(s)$. Let s such that $\widehat{P}(s) \neq E[d \mid s]$
   - If $\widehat{P}(s) < E[d \mid s]$, the informed agent buys an infinite amount at this price when he gets signal s, so the market does not clear: a contradiction
   - Similar argument if $\widehat{P}(s) > E[d \mid s]$

Next

- What if every agent has information?
- Does the market accurately aggregate information dispersed among agents?

Road Map

Rational Expectations Equilibrium

Market Efficiency

Informational Efficiency

- Definition: A market is strong form informationally efficient if prices reflect all public and private information

  $\Leftrightarrow$ Trading profit can't be predicted using all info: $E[d - p \mid \text{all info}] = 0$

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Informational Efficiency

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- Definition: A market is semi-strong form informationally efficient if prices reflect all public information

  $\Leftrightarrow$ Trading profit can't be predicted using public info: $E[d - p | \text{public info}] = 0$

- In the REE studied earlier, the market is strong form efficient

  Proof: All info $= \{s, p\}$

$$
\begin{aligned}
E[d - p \,|\, s, p] &= E[d - p \,|\, s] && \text{because } p \text{ can be backed out from } s \\
&= E[d - E[d|s] \,|\, s] \\
&= 0 && \text{Bayes law}
\end{aligned}
$$

Informational Efficiency

- Consider the same framework as before + public information
- Public signal $s_{pub} = d + \eta$ where $\eta \sim \mathcal{N}\left(0, \sigma_\eta^2\right)$ and η is orthogonal to d and s
  - Agent $i = 1$ observe s_{pub} and s
  - Agents $i = 2, \ldots, N$ observe s_{pub}

Q1. If the market is strong form efficient, then $p =$

$E[d|s]$ $\quad$ $E[d|s, s_{pub}]$ $\quad$ $E[d|s_{pub}]$

Informational Efficiency

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Q1. If the market is strong form efficient, then $p = E[d \mid s_{pub}, s]$

Q2. If $p = E[d \mid s_{pub}]$, then the market is semi-strong form efficient?

Informational Efficiency

- Consider the same framework as before + public information
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Q1. If the market is strong form efficient, then $p = E[d \mid s_{pub}, s]$

Q2. If $p = E[d \mid s_{pub}]$, then the market is semi-strong form efficient? yes

Equilibrium

- A REE of this economy is a price function $P(s_{pub}, s)$ which maps public and private signals to the price, such that $p = P(s_{pub}, s)$ is an equilibrium when agents have rational expectation

- The unique REE is $P(s_{pub}, s) =$

Equilibrium

- A REE of this economy is a price function $P(s_{pub}, s)$ which maps public and private signals to the price, such that $p = P(s_{pub}, s)$ is an equilibrium when agents have rational expectation

- The unique REE is $P(s_{pub}, s) = E[d \mid s_{pub}, s]$

- In this equilibrium, the market is strong form efficient and agents do not trade

Dispersed Information

- What if each agent has different information: does the market accurately aggregate dispersed information?

Many Informed Agents

- Each agent i receives a private signal s_i about d
- All agents observe the public signal s_{pub}
- Theorem:
  1. $P(s_{pub}, s_1, \ldots, s_N) =$

Many Informed Agents

- Each agent i receives a private signal s_i about d
- All agents observe the public signal s_{pub}
- Theorem:
  1. $P(s_{pub}, s_1, \ldots, s_N) = E\left[d \mid s_{pub}, s_1, \ldots, s_N\right]$ is a REE
  2. The market is strong form efficient
  3. Agents do not trade
- Proof: similar to the case with one informed agent

Intermediate Summary

- We studied how people trade under asymmetric information, and the properties of the ensuing market equilibrium

  1. The price reveals all public and private information

  2. Agents don't trade in equilibrium

- This raises new puzzles

  - What are the incentives to acquires information if one cannot benefit from it? If no one has incentives to acquire information, how can markets be informative?

  - Point 2. is counterfactual

Puzzle 1: Incentives to Acquire Information

- Suppose no public information and agents $i = 2, \ldots, N$ uninformed
- At date $t = -1$ agent $i = 1$ decides whether to acquire information
  - Can incur cost $C > 0$ to receive signal s at date 0

    (e.g., hire financial analysts, quants, buy data)
  - Or incur no cost and remain uninformed

Q. The agent acquires info never / if C is not too large / always?

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    (e.g., hire financial analysts, quants, buy data)
  - Or incur no cost and remain uninformed
- The agent has no incentive to acquire private information

  because the price reveals the private information, so one cannot profitably trade on private information (formal proof below)
- But then, how can markets be informative? Grossman-Stiglitz paradox (1980)

Proof

- If the agent does not acquire private information
  - Equilibrium at date 0

$$p = E[d]$$

$$E[c_1] = \overline{n}_1 p$$

  - Expected utility: $\overline{n}_1 E[d]$

- If the agent acquires private information
  - Equilibrium at date 0 for given s

$$p = E[d \mid s]$$

$$E[c_1 \mid s] = \overline{n}_1 p$$

  - Expected utility: $E[\overline{n}_1 E[d \mid s]] - C = \overline{n}_1 E[d] - C$

Puzzle 2: Trading Volume

- Trading volume in financial markets is large

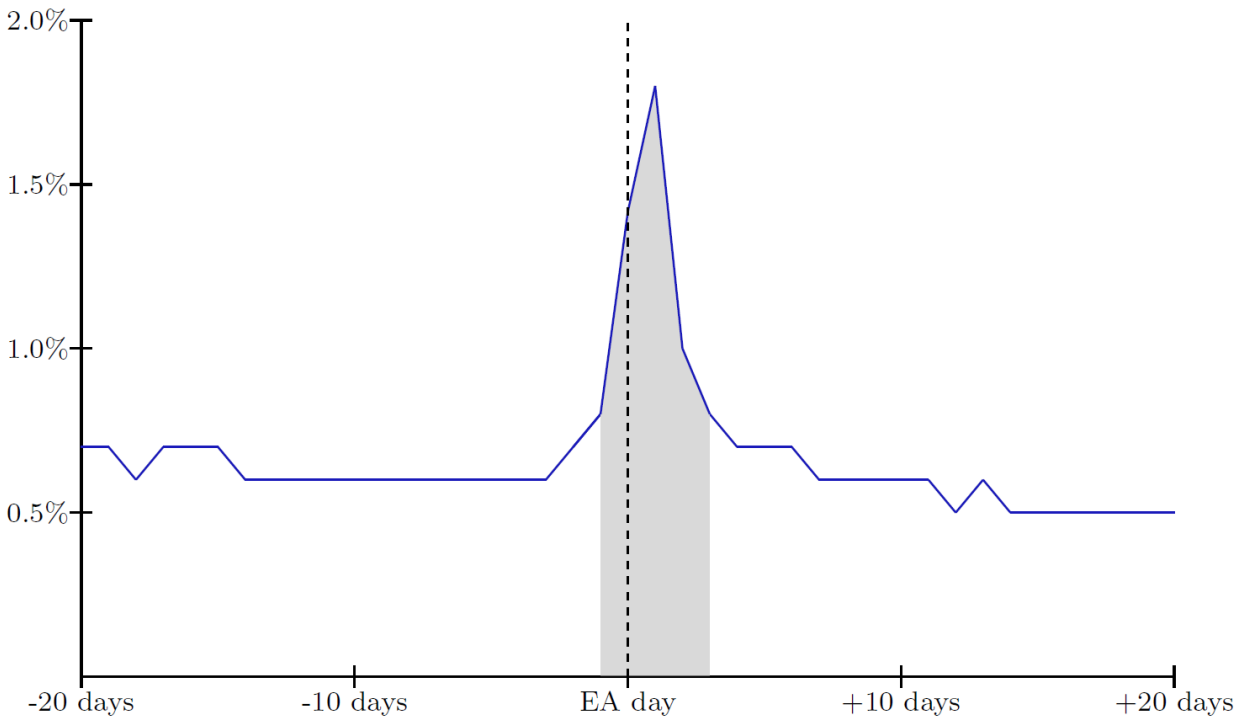
  Stock market turnover (% per year) = value of shares traded divided by market capitalization (source: World Bank)

$\approx$ 100% annual turnover

Puzzle 2: Trading Volume

- Trading volume peaks upon public information releases

Stock turnover around earnings announcements



Trading volume is computed as the percentage of shares that are turned over on each day, from 20 days preceding the announcement to 20 days after the announcement.

Solution?

Both puzzles are rooted in the inability to profitably trade on private information

Q. What is missing from our framework and may explain why trading volume is nonzero and trading on information can be profitable?

Solution?

Both puzzles are rooted in the inability to profitably trade on private information

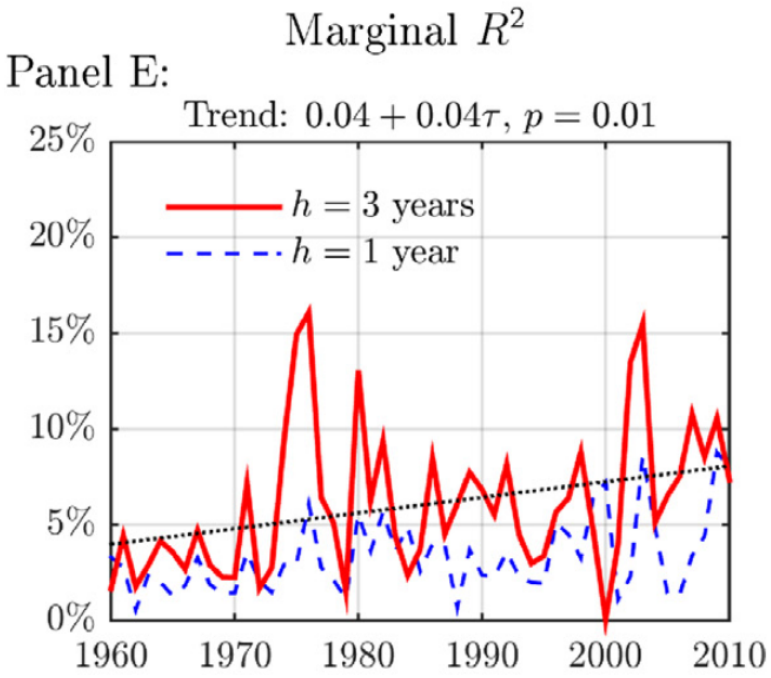
Q. What is missing from our framework and may explain why trading volume is nonzero and trading on information can be profitable?

A good introduction: Foucault, Pagano, and Roell, 2024, *Market Liquidity: Theory, Evidence, and Policy,* Oxford University Press

Bonus Slides

Have Financial Markets Become More Informative?

- Bai, Philippon and Savov (2016): measure market informativeness based on the predictive power of stock prices for future earnings[1]



[1] Bai, Philippon and Savov 2016. Have financial markets become more informative? *Journal of Financial Economics* [pdf]

Some Research Topics

- How (in)efficient are financial markets and why?
- Impact of AI, passive investing, market power, regulation
- Behavioral factors and beliefs
- Impact of financial markets on the real economy